

140L Lecture Palette

Four content colours that survive colour-vision deficiency and greyscale rendering

THE FOUR



ink

#111111

18.9:1 on white
luminance 0.006

body text, titles, template DNA



blue

#004373

10.2:1 on white
luminance 0.053

newly synthesised strand



vermillion

#BA3A13

5.7:1 on white
luminance 0.135

callouts, errors, the thing to look at



amber

#A99011

3.1:1 on white
luminance 0.284

fourth channel: RNA, displaced strand

WHY THESE, AND NOT SOME OTHER FOUR

WCAG does not recommend colours. It specifies contrast RATIOS and is entirely hue-agnostic: 4.5:1 for body text, 3:1 for large text and for graphical objects such as diagram strokes. Which hues to pick is a separate question, governed by colour-vision deficiency, which affects roughly 8% of men.

Under dichromacy you effectively have ONE hue axis, blue against yellow, plus lightness. Red, orange, yellow and green all collapse onto the yellow pole. So hue alone cannot separate four colours — lightness has to do the work. That is fortunate, because lightness is also exactly what greyscale printing preserves. Solving for one solves the other.

Black sits at 18.9:1 against white, and the 3:1 floor for graphical objects caps the lightest usable colour. Four values spaced evenly in LOG luminance across that span give the largest achievable minimum separation — a ratio of about 1.81 between each neighbour. Hue was then chosen per step to reinforce the lightness difference, and the brightness solved numerically to land on each target luminance.

PAIRWISE GREYSCALE CONTRAST

	ink	blue	vermillion	amber
ink	—	1.84:1	3.33:1	6.01:1
blue	—	—	1.81:1	3.26:1
vermillion	—	—	—	1.81:1
amber	—	—	—	—

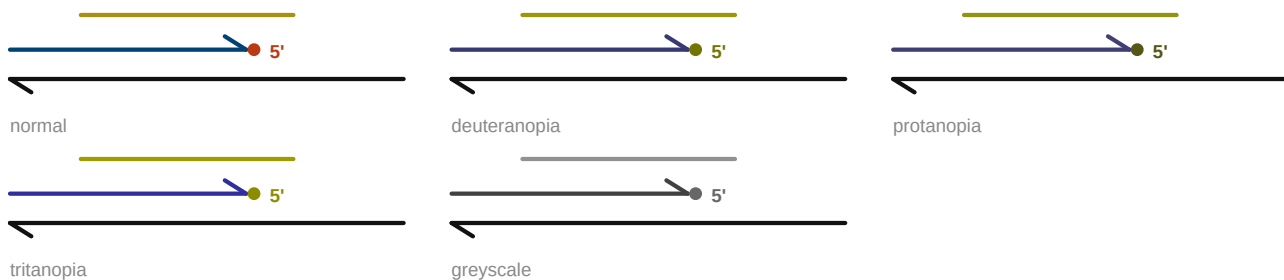
Every pair, not merely adjacent ones, clears 1.81:1 in greyscale.

Proof

The same four colours, as five different viewers see them

normal				
deuteranopia ~6% of men				
protanopia ~2% of men				
tritanopia very rare				
greyscale print / projector				

THE SAME DIAGRAM, IN EACH CONDITION



KNOWN LIMITATION

Vermillion and amber both sit on the yellow pole. To a viewer with red-green colour-vision deficiency they differ ONLY in lightness — 1.81:1, which is perceptible, but it is not a hue difference. Never let those two alone carry a distinction that matters. This is not a defect of the palette; it is a property of dichromacy, and it is why WCAG 1.4.1 requires that colour never be the sole means of conveying information. Reinforce with position, label, line style, or the written description channel the deck already carries.

Specified and implemented in `style/palette.css`. Regenerate this sheet with `python _build_palette_pdf.py`